Abstract

This project, **Road Accident Data Analysis & Visualization**, demonstrates how structured accident records can be cleaned, explored and converted into visual insights. The supplied cleaned dataset contains 15 accident records and 14 fields, including date, time, city, coordinates, weather, road condition, vehicle type, vehicles involved, casualties, severity and cause. Exploratory analysis is presented using a severity-count bar chart, a city-wise casualty chart and a correlation heatmap. The project presentation also proposes an interactive dashboard, location map, filters, CSV export and a simple risk simulator.

The dataset is a small sample for demonstrating the workflow. Therefore, the results describe only these supplied records and should not be treated as representative statistics for all road accidents.

1. Introduction

Road accidents can cause injuries, fatalities, property damage and disruption to communities. Accident data often appears as rows and columns, which can make patterns across locations, road conditions, weather and severity difficult to identify quickly.

Data analysis and visualization help transform raw records into summaries that are easier to interpret. This project uses a repeatable preprocessing workflow and charts to support exploration of accident severity, casualties by city and relationships among selected numeric variables.

2. Problem Statement

Raw accident records may contain missing values, inconsistent category labels, duplicate entries or incorrectly formatted dates. Without cleaning and visualization, it is difficult to compare accident severity, identify locations with higher recorded casualties, or examine relationships between variables. A clear and interactive analysis workflow is needed to make the data easier to explore.

3. Objectives

- Inspect and clean the supplied road accident dataset.
- Standardize labels and validate important fields such as date, time and geographic coordinates.
- Summarize accident severity and casualties across cities.
- Examine correlations among latitude, longitude, vehicles involved and casualties.
- Present findings through understandable charts and outline an interactive dashboard for further exploration.

4. Dataset Description

The cleaned workbook contains 15 rows and 14 columns. Important fields include Accident_ID, Date, Time, City, Road, Latitude, Longitude, Weather, Road_Condition, Vehicle_Type, Vehicles_Involved, Casualties, Severity and Cause.

The sample records contain 15 accidents, 36 recorded casualties and 33 vehicles involved in total. The average recorded casualties per accident is 2.40. These are sample totals, not city-wide or regional estimates.

Measure	Value
Accident records	15
Columns	14
Recorded casualties	36
Vehicles involved (sum)	33
Average casualties per record	2.40

5. Methodology

Step 1 – Load and inspect: Read the Excel dataset with Pandas and inspect dimensions, column types, summary statistics and missing values.

Step 2 – Clean: Check repeated Accident_ID values, standardize text labels, handle missing values and verify numeric and date/time fields. Missing coordinates should be checked before mapping; filled values should be documented rather than assumed to be real observations.

Step 3 – Explore: Group records by severity and city, calculate casualty totals and inspect numeric correlations.

Step 4 – Visualize: Use bar charts and a heatmap to communicate patterns. The project presentation additionally outlines an interactive map and dashboard.

Step 5 – Interpret: Describe the observed sample patterns, clearly separating actual dataset results from illustrative dashboard examples.

6. Tools and Technologies

Python: Main programming language for data analysis.
Pandas: Loading, cleaning, grouping and summarizing records.
Matplotlib and Seaborn: Producing charts and the correlation heatmap.
Streamlit: Proposed framework for an interactive web dashboard.
Plotly: Proposed interactive charts.
Folium: Proposed geographic map using latitude and longitude.

7. Results and Visualization

Accident severity: The supplied severity chart shows 10 Minor, 4 Serious and 1 Fatal records. Minor incidents form the largest category in this sample, while one record is classified as Fatal.

Casualties by city: In the supplied chart, Madurai has the largest recorded casualty total (13), followed by Salem (8), Coimbatore (6), Trichy (5) and Chennai (4). These totals match the cleaned workbook. Madurai is therefore the city with the highest casualty count in this 15-record sample.

Correlation heatmap: The supplied heatmap displays a moderate positive correlation between Latitude and Longitude (0.40), Latitude and Casualties (0.35), and Longitude and Casualties (0.37). Vehicles_Involved and Casualties show a near-zero correlation (-0.02) in the displayed matrix. Correlation describes association, not causation; with only 15 records, these values are unstable and should be interpreted cautiously.

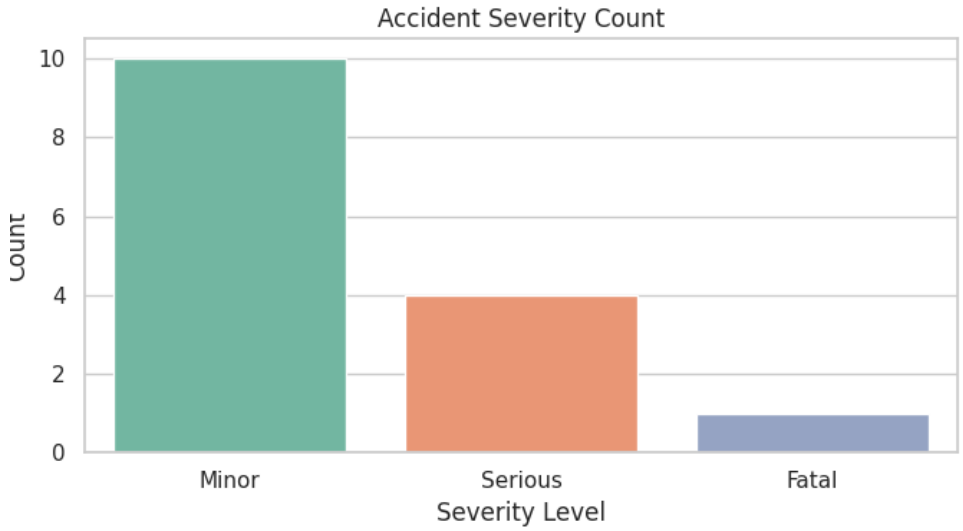


Figure 1. Accident severity count

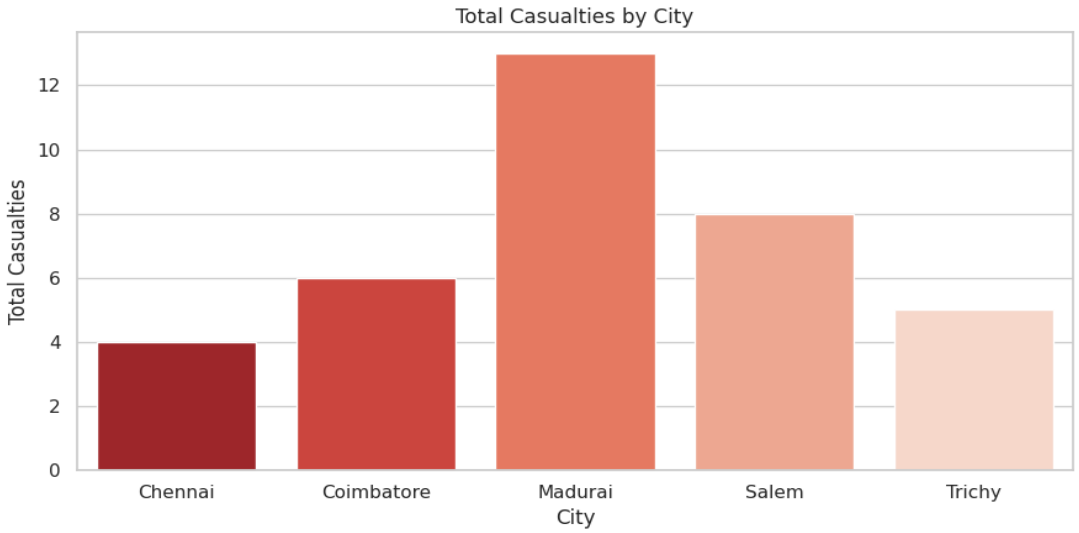


Figure 2. Total casualties by city

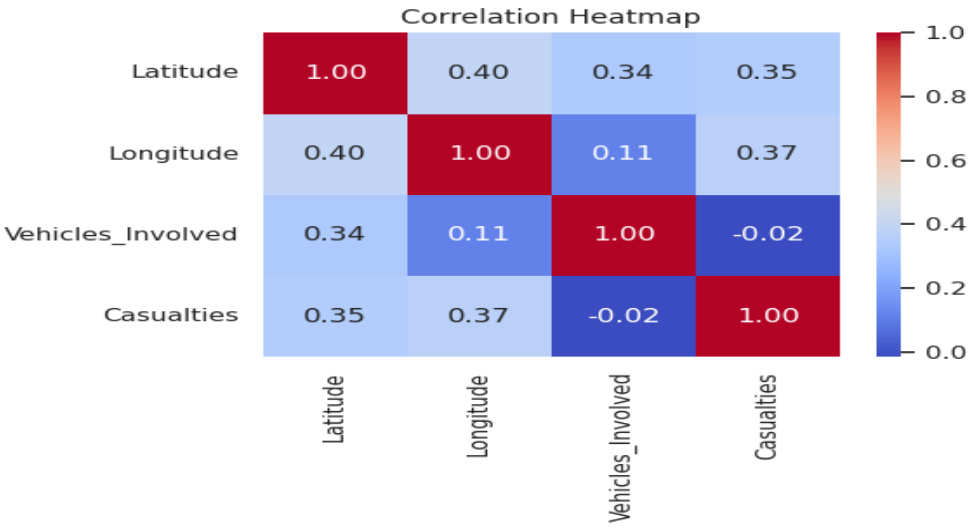


Figure 3. Correlation heatmap

8. Key Findings

• Minor severity is the most common category in the sample (10 of 15 records).
• Madurai has the highest casualty total in the supplied city-wise chart (13).
• The displayed correlation matrix does not show a meaningful linear relationship between Vehicles_Involved and Casualties for this sample.
• Location coordinates enable map-based exploration, provided that the coordinates are accurate and valid.
• A larger, verified dataset is needed before making general road-safety conclusions.

9. Proposed Dashboard Features

The project presentation proposes six features: an accident analysis dashboard, a risk simulator, an accident location map, severity analysis, interactive filters and CSV download. Filters can include city, cause and severity; map markers can be coloured by severity; and charts can update according to selected records.

The risk simulator shown in the presentation uses illustrative point weights for weather, road condition and time of day. Its score is a demonstration only—not a validated prediction. For operational use, the scoring method should be trained or calibrated with sufficient historical data and evaluated before it informs decisions.

10. Limitations

• The supplied dataset contains only 15 records, which is too small for reliable population-level conclusions.
• The sample may not represent all cities, roads, seasons or types of accidents.
• Correlation values can change substantially when a few records are added or removed.
• A risk score based on manually assigned weights is illustrative and is not a machine-learning prediction.
• The results depend on the correctness of reported accident details and geographic coordinates.

11. Future Scope

Future improvements include collecting a larger verified dataset, integrating authorized real-time or historical feeds, adding machine-learning models for severity analysis, evaluating model performance, improving map-based hotspot analysis, enabling alerts and developing a mobile-friendly interface. Any predictive feature should include validation, clear uncertainty and responsible use.

12. Conclusion

This project demonstrates a basic end-to-end data analytics workflow for road accident records: loading data, cleaning it, summarizing severity and casualties, and visualizing relationships. In the supplied sample, 10 records are Minor, 4 are Serious and 1 is Fatal; Madurai has the highest recorded casualties among the listed cities. These findings are useful for demonstrating analysis, but the small sample limits generalization. A larger, validated dataset and tested analytical methods would be necessary for practical road-safety planning.

References / Project Materials

1. Road_Accident_Cleaned_15Rows.xlsx – cleaned sample data and cleaning log.
2. Road_Accident_Messy_15Rows.xlsx – raw sample data used for comparison.
3. Untitled10 (1).ipynb – supplied Jupyter/Colab notebook for data inspection and visualization.
4. DOC-20261017-WA0003.pptx – Road Accident Data Analysis & Visualization project presentation.
5. Supplied charts: Accident Severity Count, Total Casualties by City and Correlation Heatmap.